

AI software generally falls into four distinct categories, which can be easily understood through the analogy of running a restaurant kitchen. The core differences come down to what the AI remembers, how independently it acts, and whether it becomes more or less valuable as AI technology improves.

1. Frontier Chat (The Off-Duty Master Chef)

- **What it is:** Typing directly into an AI chatbot like ChatGPT or Claude.
- **Kitchen Analogy:** An off-duty master chef sitting at your table. They can recite a brilliant five-course menu from memory, but they have no idea what is inside your fridge, whether your guest has a peanut allergy, or what you cooked the last time they visited.
- **When AI improves: Gets better.** You automatically get smarter reasoning and better answers for free, but the AI still doesn't hold onto your private context or team history.

2. Prompt Wrapper (A Website Built in Front of the Chef)

- **What it is:** A simple app or web form that takes your basic inputs and feeds them into a hidden, pre-written prompt behind the scenes.
- **Kitchen Analogy:** A website form where you select "Italian, 4 people, 1 vegetarian." It takes your answers and sends a pre-formatted message to the chef on your behalf.
- **When AI improves: Gets worse / becomes useless.** As standard AI chatbots get better at understanding regular speech, users no longer need middleman apps—they will just talk directly to the AI model instead.

3. Agent (A Line Cook Handed the Kitchen)

- **What it is:** An AI assigned a specific task that can work on its own, use external tools, check its own work, and correct its mistakes before handing back a final result.
- **Kitchen Analogy:** A line cook who checks the pantry, notices missing garlic, runs to the store, cooks the sauce, tastes it, adds salt, tastes it again, and serves it when it's right—all without you standing over them.
- **When AI improves: Gets better.** The AI's internal reasoning improves over time while your explicit safety checks, schemas, and rules stay intact.

4. Agentic System (The Entire Restaurant)

- **What it is:** An interconnected network of AI tasks running in the background based on shared memory, past data, and organizational history.
- **Kitchen Analogy:** The whole restaurant operating over a full season across multiple teams and kitchens. If a specific dish keeps getting sent back untouched across multiple waiters, the system flags it and alters tomorrow's prep sheet automatically.
- **When AI improves: Compounds in value.** Better AI reasoning combined with weeks or months of saved organizational memory makes the entire system exponentially more useful over time.

The Shift: One-Way Assembly Lines vs. Smart Loops

The Shift: One-Way Pipelines vs. Smart Loops

Older AI applications operate like a one-way assembly line: an input goes in, the AI makes a single pass, and an output comes out. If the AI makes a mistake, it cannot go backward to double-check itself, and closing the browser tab wipes its memory completely.

Modern AI systems introduce a backwards loop—the AI cook tastes the soup, adjusts the recipe, and tastes it again until it is right. Once an AI can loop and re-check its own work, four requirements become essential:

- **1. Saved State (The Restaurant Ledger):** Memory lives outside the conversation in a database, not in the chat window. If the shift changes, the pantry logs and guest allergy notes remain intact.
- **2. Hard Loop Limits (The Kitchen Timer):** Every loop must have a strict maximum limit. Otherwise, an AI trying endlessly to perfect an output will run forever and generate massive cloud bills.
- **3. Explicit Contracts (The Printed Order Ticket):** Information passed between steps must follow strict rules and verified sources—like a printed table order—rather than vague, unverified text passed between steps.
- **4. Human Approval (The Head Chef Sign-Off):** Pausing for a human isn't a crash; it's a built-in checkpoint. The workflow pauses, the server sends a request to the manager's phone, and the system resumes only after sign-off.